SPATIO-TEMPORAL ADAPTIVE PATCH SIZES FOR EFFICIENT DIFFUSION TRANSFORMERS

Team: CLIP that Chat
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1 PROBLEM STATEMENT

Diffusion transformers (DiTs) spend the same compute on every token at every denoising step, regardless of how much detail each region requires. Adaptive Patch Transformers (APTs) show Vision Transformers (ViTs) can use larger patches in low-entropy regions without losing accuracy (Choudhury et al., 2025), and DDiT shows a pretrained DiT can use coarser patches in early denoising steps (Kim et al., 2026). Building on these findings, we will investigate whether adapting patch sizes across both space and time can reduce sampling compute without compromising sample quality. Our main idea is to apply APT's spatially adaptive patch sizes to diffusion transformers (H1), and then combine this with DDiT's timestep-dependent patch schedule (H2). We hypothesize that detail is uneven in both space and time, so adapting along both axes beats either alone. At approximately matched sample quality, we ask:

1. *Within-step adaptivity (H1).* At matched sampling FLOPs, does spatially mixed patching within a step beat DDiT-style uniform-per-step patching?
2. *Joint adaptivity (H2).* Does a joint space-and-time schedule give a better quality–compute Pareto frontier than spatial-only or timestep-only schedules?
3. *Selection signal.* Which per-region score best predicts where fine patches matter, measured by a rank-based *allocation gap* between score and quality lost when coarsening?

2 MOTIVATION

Attention cost scales quadratically with respect to the token count $N = HW/p^2$ and is paid at each of the $\sim$50 sampling steps, so doubling the patch size ($4\times$ fewer tokens) gives large per-step speedups (Kim et al., 2026). If this can be made adaptive with light LoRA fine-tuning of a pretrained model (Hu et al., 2022), sampling gets cheaper without retraining, which matters at academic compute scale. Efficiency *is* the claim here, so we report FLOPs and latency, as FLOP savings can overstate real gains. A negative result would show whether spatial adaptivity adds anything beyond DDiT's per-step schedule.

3 RELATED WORK

APT (Choudhury et al., 2025) picks patch sizes from image entropy for ViTs, improving ViT-L throughput by about +40% throughput with no downstream loss, but targets recognition on clean inputs. DDiT (Kim et al., 2026) adds LoRA-adapted patch-embedding layers to a pretrained DiT and a training-free scheduler based on third-order finite differences of the latent, reporting about $2.2\times$ speedup on FLUX-1.Dev ($3.52\times$ with caching); it uses one patch size per step and names within-step variation as future work. Related ideas include multi-patch-size training (Beyer et al., 2023), mixed-resolution ViTs (Ronen et al., 2023), and token-density control in DiTs (Chang et al., 2024); caching (Liu et al., 2025) is complementary. **How we differ:** we do not claim novelty for per-step patch scheduling or for spatially mixed patches in ViTs; our contribution is a matched test of spatial, temporal, and joint adaptivity on one diffusion backbone. Since APT's pixel-entropy criterion is undefined on noisy latents, we also compare diffusion-appropriate selection signals.

4 Approach

Method. Start from a pretrained DiT-XL/2 (base patch 2 on the 32×32 latent; 256 tokens). Partition the latent into B blocks of $g\times g$ latent pixels (g=8). At step t, block b is tokenized with patch size $p_{b,t} \in \{2, 4, 8\}$, sharing one token sequence per image:

$$N_t = \sum_{b=1}^{B} \left(\frac{g}{p_{b,t}}\right)^2, \qquad p_{b,t} = \max\{p : s_{b,t}^{(p)} < \tau(t)\} \ \text{(default 2)},$$

where $s_{b,t}^{(p)}$ is the ρ-th percentile, over the $p\times p$ patches in block b, of the within-patch standard deviation of the latent's third-order difference along the sampling trajectory (DDiT's signal, computed per block instead of averaged per step). The knob $\tau(t)$ sets the compute budget. Following DDiT, we add per-size patch (de-)embedding layers, a patch-size embedding, interpolated positional embeddings, and rank-32 LoRA. Training matches the frozen full-resolution model's noise prediction, $\mathcal{L} = \|\epsilon_{\theta_L}(z_t^{\text{mixed}}, t) - \epsilon_{\theta_T}(z_t^2, t)\|^2$, on about 50k samples generated by the base model (no external data), with random spatially correlated layouts and timesteps.

Signal comparison. We compare (i) the per-block score above, (ii) APT-style entropy of the predicted clean latent $\hat{x}_0$ (defined at all timesteps, unlike entropy of the noisy input), and (iii) a timestep-only control (all blocks share p_t), and ablate the scale set, g, and τ. We repeat the best configuration on PixArt-α (Chen et al., 2024) for text-to-image.

Stretch goal (if time permits). Extend to a video DiT with a lightweight recurrent module that carries patch decisions across frames, so the scheduler is not recomputed from scratch for every frame.

5 Evaluation

Quality. Class-conditional ImageNet-256 (FID, sFID, IS, precision/recall; FID-10k for ablations, FID-50k for final runs) and, on PixArt-α, COCO (FID, CLIP score, ImageReward), plus SSIM/LPIPS against the base model's outputs. **Compute.** Sampling GFLOPs (including the N^2 attention term and scheduler), average tokens per step, and latency on an RTX 5090. **Baselines.** Full-token DiT (50 steps), fewer steps at matched FLOPs, fixed p=4, and our re-implementation of DDiT's per-step schedule on the same backbone.

Success criteria. "Match" means FID within δ of the full-token model, with δ fixed before the main runs from its seed-to-seed spread (to be measured); smaller differences are ties. H1 is supported if spatial mixing has lower FID than uniform-per-step patching, beyond δ, at three or more matched FLOP budgets; H2 if the joint schedule dominates both spatial-only and timestep-only schedules. The allocation gap is $(1 - \rho_S)/2$, with ρ_S the Spearman correlation between a block's mean score and the LPIPS increase when only that block is forced to p=8 (500 samples). Equally informative outcomes: no gain over DDiT-style scheduling, FLOP savings that vanish in latency, or a timestep-only schedule matching the per-block variants.

6 Resources

Compute. The team has three RTX 5090s (32 GB, across two members) and we assume we can get 80 PARCC GPU-hours in total. We assume about 100 TFLOP/s effective and 1.25–1.5$\times$ mixed-attention overhead (optimistic; to be re-measured in week one). **Data and models.** Public benchmarks (ImageNet reference statistics, COCO captions) plus base-model samples; DiT-XL/2 ($\sim$675M) and PixArt-α ($\sim$0.6B), not FLUX-1.Dev (12B), which is too large to fine-tune within budget. Results are small-scale.

Stage	Hours	Where	Contents
Baselines, distillation	18	5090	full-token/fewer-step/fixed-p baselines; mixed-scale LoRA
Scheduler ablations	20	5090	signals, scales, g, τ sweeps at FID-10k
Final ImageNet sweeps	25	PARCC	FID-50k, all arms, 2 sampling seeds
PixArt-α + COCO	30	PARCC	distillation and evaluation
Reserve, video stretch	20	PARCC	reruns; video with recurrent module if time

Timeline and fallback. Midpoint (Week 9): mixed-scale tokenization, distillation, and baselines working. Draft report (Week 12): scheduler comparison and ablations. If compute runs short, we drop PixArt-α and the video stretch goal first, preserving the ImageNet tests of H1 and H2.

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